

Slide 1:

Good afternoon one and all! I am Hrishikesh Gawas, pursuing my Bachelor's degree in Electronics and Communication Engineering from PES University, Bangalore. Today, I will be presenting about my paper titled "Design and Evaluation of an Early Fire Detector for Human Space Capsule", co-authored by Badrisha, Research assistant (CORI), Dr. Manikandan, professor at Dept. of ECE & CORI, PES University. The proposed work is an outcome from collaboration with SAC ISRO, under RESPOND program. I would like to thank Akash Gupta and Payal Sharma from ISRO, Ahmedabad for their support who are also the co-authors of this paper.

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Presence of an electric spark or smoke is considered an early indication for fire. In this paper, design and evaluation of two variants of early fire detector for human space capsule are presented.

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Most mission experiences reported early fire detection by humans through smell, rather than onboard sensors. A detailed review on past fire accidents in space exploration is reported in [1]. Fire detection and suppression systems used in the international space station are reported in [3], [4] & [5]. Use of a network of autonomous robots for detection of smoke, fire and toxic chemical residue is proposed in [10].

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The proposed system is capable of detecting smoke comprising of volatile organic compounds (VOCs) produced due to overheating of materials used on board human space capsule before the fire begins. It can also detect other early signs of fire such as electric sparks and overheating, including invisible flame. Both the variants employ a combination of E-Nose and E-Eye, thus making the early fire detection system more robust in detection of VOCs.

(Explain Diagrams)

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After a detailed study and comparison between various sensors, ION Science Ltd. Make MiniPID2 Wide Range 10.6eV Photo Ionization Detector (PID) gas sensor, capable of detecting 1000+ VOCs was finalized. The detector is also equipped with MLX90640 thermal camera which has a viewing angle of 110x75 degrees. The MINIPID sensor provides analog voltage values which are amplified and sampled through inbuilt ADC on ESP32 series microcontroller. A display and Buzzer are provided for quick audio-visual stimulus. The microcontroller also creates a local online dashboard to view real-time results.

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(Explain Diagrams)

Slide 7:

In order to assess the efficacy of the proposed early fire detection system, several experiments were carried out for the E-Eye and E-Nose part of the system.

(Explain Diagrams)

Slide 8:

To assess the performance of the E-Nose part of the system, VOC gases produced due to overheating and burning of materials used in the human space capsule were detected and quantified. Six materials heavily used in space capsules susceptible to fire were chosen. These were heated to observe a change of 20mV in sensor output equivalent to 50 PPM. The change in weight of materials was noted.

(Explain Diagrams)

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In conclusion, this paper proposes the design of an early fire detector which is supported by exhaustive experimental analysis specific to human space capsule. It mainly comprises of E-Nose & E-Eye and may be constructed as shown in two variants. A different approach has been proposed by using a VOC sensor to improve variety of gases being detected lowering the response time to detect fire and is complemented by the redundancy made by integrating two sensors.

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These are the references we used as discussed earlier. Thank you everyone!